

UA1 Projet1 G4 Les_Hadoop_Riders

Projet 1 — Déploiement et Exploitation d'un Cluster Hadoop avec Docker

Cours : Bases de données massives avancées (IFM30522)

Groupe : G4 — Les_Hadoop_Riders

Étudiant

Komla Petro Asinyo	Kassoum Dene
Joel Kazoni Tugirimana	Forbes Magène
Frank A Simo Ngounou	Wren Surprenant-Nicolson

Docker Hub : <https://hub.docker.com/r/leshadoopridders/hadoop-tp-g4>

Image : leshadoopridders/hadoop-tp-g4:1.0

Code source : https://github.com/asinyopetro/Les_Hadoop_Riders_G4

Partie 1 — Contexte théorique et architecture (5 pts)

1. Cas d'usage Big Data (1.5 pt)

Nous avons choisi le secteur de l'**e-commerce**.

Exemple : une chaîne de magasins qui garde les tickets de caisse, les logs du site et les mouvements de stock. Chaque jour le volume grossit, les données arrivent souvent, et les formats changent (CSV, JSON, texte, images produits).

Les **3V** dans ce cas :

- **Volume** : beaucoup d'événements, historique sur plusieurs années.
- **Vélocité** : commandes et paiements en continu ; il faut pouvoir réagir vite (fraude, rupture de stock).
- **Variété** : tables, logs, parfois des médias.

HDFS sert au stockage distribué. YARN sert à lancer les jobs (MapReduce, etc.) sur le cluster.

2. HDFS vs système de fichiers classique (1.5 pt)

Sur un système de fichiers classique (ext4, NTFS), un fichier vit sur **une** machine. C'est l'OS local qui gère les métadonnées et le contenu.

Avec **HDFS**, le fichier est coupé en **blocs**. Les blocs sont placés et **répliqués** sur plusieurs DataNodes. Le **NameNode** garde le namespace (noms, dossiers, où sont les blocs).

En pratique : le FS local est centralisé ; HDFS est distribué et plus tolérant aux pannes grâce à la réplication.

3. Rôle de YARN — ResourceManager et NodeManager (2 pts)

Quand on soumet une application :

- Le **ResourceManager** (master) reçoit la demande, voit les ressources du cluster et alloue des conteneurs. Il lance aussi l'ApplicationMaster.
- Le **NodeManager** (chaque worker) démarre et surveille les conteneurs **sur son nœud**, puis remonte l'état au ResourceManager.

Donc : le ResourceManager orchestre ; le NodeManager exécute sur la machine.

Partie 2 — Déploiement et manipulation pratique (15 pts)

A. Mise en place de l'environnement Docker (4 pts)

1. Déploiement NameNode + 5 DataNodes (1.5 pt)

Nous avons déployé six conteneurs :

- `hadoop-master` : NameNode, ResourceManager, SecondaryNameNode
- `hadoop-worker1` à `hadoop-worker5` : DataNode + NodeManager

Réseau Docker : `hadoop-net`

Ports exposés : **9870** (UI HDFS), **8088** (UI YARN), **9000** (HDFS RPC)

```
docker compose up --build -d
docker ps
```

```
Windows PowerShell — Les_Hadoop_Riders_G4

PS C:\Users\asiny\Desktop\Les_Hadoop_Riders_G4> docker ps
NAME                STATUS              PORTS
hadoop-worker2      Up 10 hours        8042/tcp, 8088/tcp, 9000/tcp, 9864/tcp, 9870/tcp
hadoop-worker3      Up 10 hours        8042/tcp, 8088/tcp, 9000/tcp, 9864/tcp, 9870/tcp
hadoop-worker5      Up 10 hours        8042/tcp, 8088/tcp, 9000/tcp, 9864/tcp, 9870/tcp
hadoop-worker1      Up 10 hours        8042/tcp, 8088/tcp, 9000/tcp, 9864/tcp, 9870/tcp
hadoop-worker4      Up 10 hours        8042/tcp, 8088/tcp, 9000/tcp, 9864/tcp, 9870/tcp
hadoop-master       Up 10 hours        0.0.0.0:8088->8088/tcp, [::]:8088->8088/tcp, 0.0.0.0:9000->9000/tcp, [::]:9000->9000/tcp, 0.0.0.0:9870->9870/tcp, [::]:9870->9870/tcp

PS C:\Users\asiny\Desktop\Les_Hadoop_Riders_G4> docker exec -it hadoop-master hdfs dfsadmin -report
Configured Capacity: 5405505884160 (4.92 TB)
Present Capacity: 5032864239616 (4.58 TB)
DFS Remaining: 5032862023680 (4.58 TB)
DFS Used: 2215936 (2.11 MB)
DFS Used%: 0.00%
Replicated Blocks:
  Under replicated blocks: 0
  Blocks with corrupt replicas: 0
  Missing blocks: 0
  Missing blocks (with replication factor 1): 0
  Low redundancy blocks with highest priority to recover: 0
  Pending deletion blocks: 0
Erasure Coded Block Groups:
  Low redundancy block groups: 0
  Block groups with corrupt internal blocks: 0
  Missing block groups: 0
  Low redundancy blocks with highest priority to recover: 0
  Pending deletion blocks: 0

-----
Live datanodes (5):

Name: 172.23.0.3:9866 (hadoop-worker5.les_hadoop_riders_g4_hadoop-net)
Hostname: hadoop-worker5
Decommission Status : Normal
Configured Capacity: 108110176832 (100.85 GB)
```

Capture terminal — docker ps / dfsadmin

2. Vérification des DataNodes (1.5 pt)

docker exec -it hadoop-master hdfs dfsadmin -report

Résultat observé : **Live datanodes (5)**.
Interface NameNode : <http://localhost:9870>

Hadoop

Overview

Datanodes

Datanode Volume Failures

Snapshot

Startup Progress

Utilities

Overview 'hadoop-master:9000' (✓active)

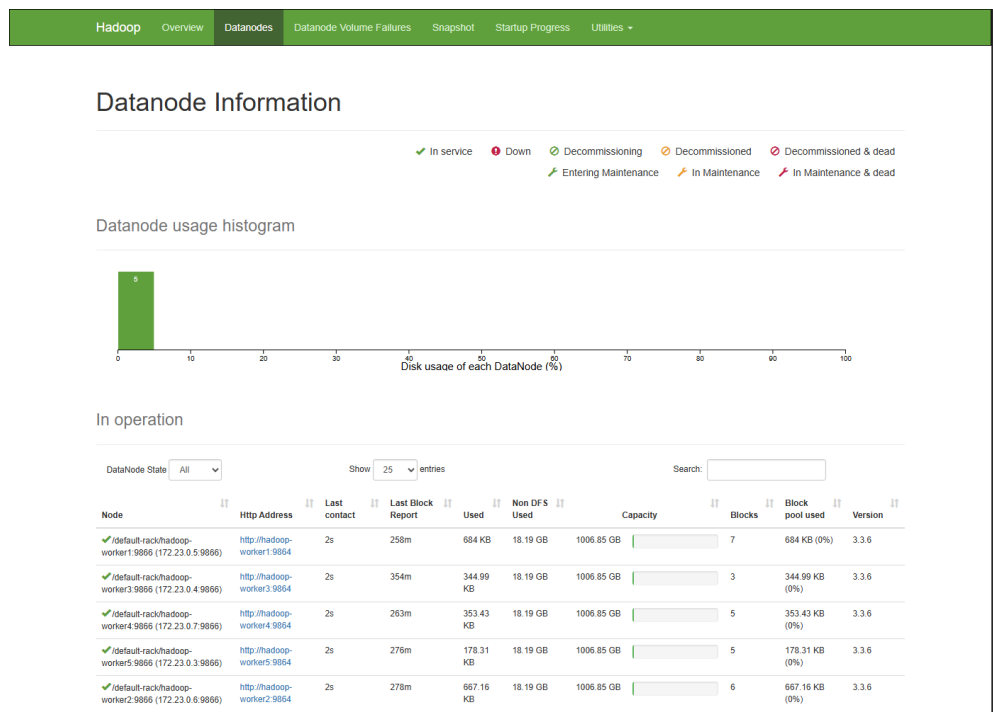
Started:	Sat Oct 03 11:02:46 -0400 2026
Version:	3.3.6, r1be78238728da9266a4f88195058f08fd012bf9c
Compiled:	Sun Jun 18 04:22:00 -0400 2023 by ubuntu from (HEAD detached at release-3.3.6-RC1)
Cluster ID:	CID-c82f423-c208-4cdf-9c10-5573319713da
Block Pool ID:	BP-329230766-172.23.0.2-1791033373712

Summary

Security is off.
Safemode is off.
32 files and directories, 9 blocks (9 replicated blocks, 0 erasure coded block groups) = 41 total filesystem object(s).
Heap Memory used 89.9 MB of 242 MB Heap Memory. Max Heap Memory is 2.4 GB.
Non Heap Memory used 69.58 MB of 72.25 MB Committed Non Heap Memory. Max Non Heap Memory is <unbounded>.

Configured Capacity:	4.92 TB
Configured Remote Capacity:	0 B
DFS Used:	2.18 MB (0%)
Non DFS Used:	90.97 GB
DFS Remaining:	4.58 TB (93.11%)
Block Pool Used:	2.18 MB (0%)
DataNodes usages% (Min/Median/Max/stdDev):	0.00% / 0.00% / 0.00% / 0.00%
Live Nodes	5 (Decommissioned: 0, In Maintenance: 0)
Dead Nodes	0 (Decommissioned: 0, In Maintenance: 0)
Decommissioning Nodes	0

UI NameNode



Liste des DataNodes

3. Publication de l'image (1 pt)

docker login

docker push leshadoopridders/hadoop-tp-g4:1.0

URL publique : <https://hub.docker.com/r/leshadoopridders/hadoop-tp-g4>

(Note : Docker Hub impose les minuscules, d'où g4 plutôt que G4.)

B. Manipulation avancée sur HDFS (6 pts)

4. Répertoires et droits (2 pts)

Fichier local créé : data/transactions.csv (colonnes ID, Date, Montant, Magasin — 10 lignes).

```
hdfs dfs -mkdir -p /data/ventes/2026
```

```
hdfs dfs -chmod 755 /data/ventes/2026
```

```
hdfs dfs -put -f /data/transactions.csv /data/ventes/2026/
```

```
hdfs dfs -chmod 644 /data/ventes/2026/transactions.csv
```

5. Transfert et vérification des blocs (2 pts)

```
hdfs fsck /data/ventes/2026/transactions.csv -files -blocks -locations
```

```
hdfs dfs -stat "%n | taille=%b | replication=%r | block_size=%o"
/data/ventes/2026/transactions.csv
```

- Résultats : - taille : 377 octets
- réplication initiale : 3
- 1 bloc (fichier petit ; taille de bloc par défaut 128 Mo)
- état FSCK : HEALTHY

```

root@hadoop-master:~# hdfs dfs -ls /data/ventes/2026
Found 2 items
-rw-r--r-- 2 hadoop supergroup 377 2026-10-03 13:30 /data/ventes/2026/transactions.csv
-rw-r--r-- 3 hadoop supergroup 567 2026-10-03 13:31 /data/ventes/2026/transactions_merged.csv
root@hadoop-master:~# hdfs dfs -cat /data/ventes/2026/transactions.csv | head -n 5
ID,Date,Montant,Magasin
T001,2026-01-05,49.99,Montreal-Centre
T002,2026-01-06,129.50,Laval
T003,2026-01-07,15.25,Quebec
T004,2026-01-08,250.00,Montreal-Ouest
root@hadoop-master:~# hdfs fsck /data/ventes/2026/transactions.csv -files -blocks -locations 2>&1 | head -n 25
Connecting to namenode via http://hadoop-master:9870/fsck?ugi=root&files=1&blocks=1&locations=1&path=%2Fdata%2Fventes%2F2026%2Ftransactions.csv
FSCK started by root (auth:SIMPLE) from /172.23.0.2 for path /data/ventes/2026/transactions.csv at Sun Oct 04 01:02:57 GMT 2026

/data/ventes/2026/transactions.csv 377 bytes, replicated: replication=2, 1 block(s): OK
0. BP-329230766-172.23.0.2-1791033373712:blk_1073741825_1001 len=377 Live_rep1=2 [DatanodeInfoWithStorage[172.23.0.4:9866,DS-6b0bc5c8-0343-4441-977a-45186fdef03,DISK], DatanodeInfoWithStorage[172.23.0.6:9866,DS-f042a204-27cf-4970-b237-39c216ef47fb,DISK]]

Status: HEALTHY
Number of data-nodes: 5
Number of racks: 1
Total dirs: 0
Total symlinks: 0

Replicated Blocks:
Total size: 377 B
Total files: 1
Total blocks (validated): 1 (avg. block size 377 B)
Minimally replicated blocks: 1 (100.0 %)
Over-replicated blocks: 0 (0.0 %)
Under-replicated blocks: 0 (0.0 %)
Mis-replicated blocks: 0 (0.0 %)
Default replication factor: 3
Average block replication: 2.0
Missing blocks: 0
Corrupt blocks: 0

```

Capture terminal — HDFS

6. Lecture et concaténation (1 pt)

```

hdfs dfs -cat /data/ventes/2026/transactions.csv | head -n 5
hdfs dfs -put -f /data/transactions_part2.csv /data/ventes/2026/
hdfs dfs -getmerge /data/ventes/2026 /tmp/transactions_merged.csv
hdfs dfs -put -f /tmp/transactions_merged.csv
/data/ventes/2026/transactions_merged.csv

```

Les 5 premières lignes sont lues directement depuis HDFS. Les deux fichiers ont été fusionnés via getmerge.

7. Suppression / corbeille (1 pt)

```

hdfs dfs -rm /data/ventes/2026/transactions_part2.csv
hdfs dfs -D fs.trash.interval=10080 -rm /data/ventes/2026/to_delete.csv
hdfs dfs -ls -R /user/hadoop/.Trash
# suppression définitive :
# hdfs dfs -rm -skipTrash /chemin/fichier

```

Sans fs.trash.interval, la suppression peut être définitive. Avec un intervalle > 0, le fichier passe dans .Trash. L'option -skipTrash force la suppression définitive.

C. Exécution d'un job YARN (5 pts)

Exemple choisi : calcul de π (Monte Carlo).

```
yarn jar $HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-3.3.6.jar pi 4 1000
```

Résultat : - Application : application_1791039798429_0001
- Nom : QuasiMonteCarlo
- État final : **SUCCEEDED**
- Estimation de $\pi \approx 3.14$

```
root@hadoop-master — YARN

root@hadoop-master:~# yarn application -list -appStates ALL
docker : 2026-10-04 01:03:01,304 INFO client.DefaultNoHARMFaloverProxyProvider: Connecting to Resource Manager at
hadoop-master/172.23.0.2:8032
Au caractère C:\Users\sasny\AppData\Local\Temp\ps-script-67860ea5-b239-4dd4-997c-850c8a40b275.ps1:107 : 9
+ $yarn = docker exec hadoop-master bash -c 'yarn application -list -ap ...
+ ~~~~~
+ CategoryInfo          : NotSpecified: (2026-10-04 01:0...172.23.0.2:8032:String) [], RemoteException
+ FullyQualifiedErrorId : NativeCommandError

Total number of applications (application-types: [], states: [NEW, NEW_SAVING, SUBMITTED, ACCEPTED, RUNNING, FINISHED, FAILED, KILLED] and tags:
[]):1
Application-Id      Application-Name      Application-Type      User      Queue      State
-----
application_1791039798429_0001      QuasiMonteCarlo      MAPREDUCE      root      default      FINISHED
SUCCEEDED      100% http://hadoop-worker1:19888/jobhistory/job/job_1791039798429_0001
root@hadoop-master:~# yarn application -status application_1791039798429_0001 2>&1 | head -n 30
2026-10-04 01:03:05,625 INFO client.DefaultNoHARMFaloverProxyProvider: Connecting to Resource Manager at hadoop-master/172.23.0.2:8032
2026-10-04 01:03:06,463 INFO conf.Configuration: resource-types.xml not found
2026-10-04 01:03:06,464 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
Application Report :
  Application-Id : application_1791039798429_0001
  Application-Name : QuasiMonteCarlo
  Application-Type : MAPREDUCE
  User : root
  Queue : default
  Application Priority : 0
  Start-Time : 1791070566554
  Finish-Time : 1791070642461
  Progress : 100%
  State : FINISHED
  Final-State : SUCCEEDED
  Tracking-URL : http://hadoop-worker1:19888/jobhistory/job/job_1791039798429_0001
  RPC Port : 41245
  AM Host : hadoop-worker1
  Aggregate Resource Allocation : 210745 MB-seconds, 245 vcore-seconds
  Aggregate Resource Preempted : 0 MB-seconds, 0 vcore-seconds
  Log Aggregation Status : DISABLED
```

Capture terminal — YARN

FINIS

Cluster

About

Nodes

Node Labels

Applications

NEW

NEW_SAVING

SUBMITTED

ACCEPTED

RUNNING

FINISHED

FAILED

KILLED

Scheduler

Tools

Cluster Metrics

Apps Submitted	Apps Pending	Apps Running	Apps Completed	Containers Running	Used R
1	0	0	1	0	<memory:0 B, vCores

Cluster Nodes Metrics

Active Nodes	Decommissioning Nodes	Decommissioned Nodes
5	0	0

Scheduler Metrics

Scheduler Type	Scheduling Resource Type	Minimum Allocation
Capacity Scheduler	[memory-mb (unit=Mi), vcores]	<memory:256, vCores:1>

Show 20 entries

ID	User	Name	Application Type	Application Tags	Queue	Application Priority	StartTime	LaunchTime	FinishTime
application_1791039798429_0001	root	QuasiMonteCarlo	MAPREDUCE		default	0	Sat Oct 3 19:36:06 -0400 2026	Sat Oct 3 19:36:11 -0400 2026	Sat Oct 3 19:37:22 -0400 2026

Showing 1 to 1 of 1 entries

UI YARN — applications


Application application_1791039798429_0001
Logged in as: drwho

Cluster

- About
- Nodes
- Node Labels
- Applications
- NEW
- NEW SAVING
- SUBMITTED
- ACCEPTED
- RUNNING
- FINISHED
- FAILED
- KILLED
- Scheduler

Tools

Application Overview

User: root
Name: QuasiMonteCarlo
Application Type: MAPREDUCE
Application Tags:
Application Priority: 0 (Higher Integer value indicates higher priority)
YarnApplicationState: FINISHED
Queue: default
FinalStatus Reported by AM: SUCCEEDED
Started: Sat Oct 03 23:36:06 +0000 2026
Launched: Sat Oct 03 23:36:11 +0000 2026
Finished: Sat Oct 03 23:37:22 +0000 2026
Elapsed: 1mins, 15sec
Tracking URL: History
Log Aggregation Status: DISABLED
Application Timeout (Remaining Time): Unlimited
Diagnostics:
Unmanaged Application: false
Application Node Label expression: <Not set>
AM container Node Label expression: <DEFAULT_PARTITION>

Application Metrics

Total Resource Preempted: <memory:0, vCores:0>
Total Number of Non-AM Containers Preempted: 0
Total Number of AM Containers Preempted: 0
Resource Preempted from Current Attempt: <memory:0, vCores:0>
Number of Non-AM Containers Preempted from Current Attempt: 0
Aggregate Resource Allocation: 210745 MB-seconds, 245 vcore-seconds
Aggregate Preempted Resource Allocation: 0 MB-seconds, 0 vcore-seconds

Show: 20 entries

Search:

Attempt ID	Started	Node	Logs	Nodes blacklisted by the app	Nodes blacklisted by the system
appattempt_1791039798429_0001_0000001	Sat Oct 3 19:36:06 -0400 2026	http://hadoop-worker1.804f2	Logs	0	0

Showing 1 to 1 of 1 entries

First Previous 1 Next Last

Détail application SUCCEEDED

Partie 3 — Administration, monitoring et analyse (10 pts)

1. Interfaces web (4 pts)

NameNode (<http://localhost:9870>)

On voit bien les 5 DataNodes. L'espace DFS utilisé est faible (labo). Pas de missing block pendant nos tests.

YARN (<http://localhost:8088>)

Le job pi est en **FINISHED** / **SUCCEEDED**. On voit aussi la mémoire et le temps d'exécution (environ 198589 MB-seconds et 216 vcore-seconds sur un des runs).

2. Facteur de réplication (3 pts)

```
hdfs dfs -setrep -w 2 /data/ventes/2026/transactions.csv
```

Après la commande, la réplication passe à 2.

Le NameNode change la cible. Comme on passe de 3 à 2, une copie en trop est retirée. L'option -w attend que ce soit terminé.

3. Retour d'expérience / troubleshooting (3 pts)

Problème 1 — pull Docker Hub avant le build

Symptôme : pull access denied pour l'image du groupe.

Cause : Compose tentait de télécharger une image pas encore publiée.

Solution : pull_policy: never et build local de l'image.

Problème 2 — majuscules dans le tag

Symptôme : repository name must be lowercase.

Solution : tag final leshadoopridders/hadoop-tp-g4:1.0.

Problème 3 — téléchargement Hadoop trop lent

Symptôme : build bloqué longtemps sur archive.apache.org.

Solution : image de base apache/hadoop:3.3.6 + nos fichiers de configuration et entrypoint.sh.

Annexes

Structure du code source

```
Les_Hadoop_Riders_G4/
├── Dockerfile
├── docker-compose.yml
├── entrypoint.sh
├── config/
│   ├── core-site.xml
│   ├── hdfs-site.xml
│   ├── yarn-site.xml
│   ├── mapred-site.xml
│   └── workers
├── data/
│   ├── transactions.csv
│   └── transactions_part2.csv
├── scripts/
│   ├── demo-hdfs.sh
│   └── demo-yarn-pi.sh
├── captures/
└── README.md
```

Lien Docker Hub

<https://hub.docker.com/r/leshadoopridders/hadoop-tp-g4>